Lab Report 4: Investigate STP Loop Prevention

1. Objectives

The primary objectives of this laboratory exercise were to:

- Observe and describe the operation of Spanning Tree Protocol (STP).
- Explain how STP prevents switching loops while allowing redundancy in switched networks.
- Watch the spanning-tree convergence process during a topology change.

2. Network Topology & Architecture

The network consists of a redundant triangle topology containing three switches (S1, S2, and S3) interconnected via Gigabit Ethernet trunks. This physical layout provides hardware redundancy but would trigger catastrophic broadcast storms and MAC table instability without STP.

3. Reference Addressing Table

Device	Interface	IP Address	Subnet Mask	Connected To
S1	VLAN 1	192.168.1.11	255.255.255.0	Core Node
S2	VLAN 1	192.168.1.12	255.255.255.0	Distribut...
S3	VLAN 1	192.168.1.13	255.255.255.0	Distribut...
PC1	NIC	192.168.1.21	255.255.255.0	S2 (F0/1)
PC2	NIC	192.168.1.22	255.255.255.0	S3 (F0/1)

4. Switch Configurations

Switch 1 (S1)

```
None
S1> enable
S1# configure terminal
S1(config)# interface range g0/1 - 2
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.11 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
```

Switch 2 (S2)

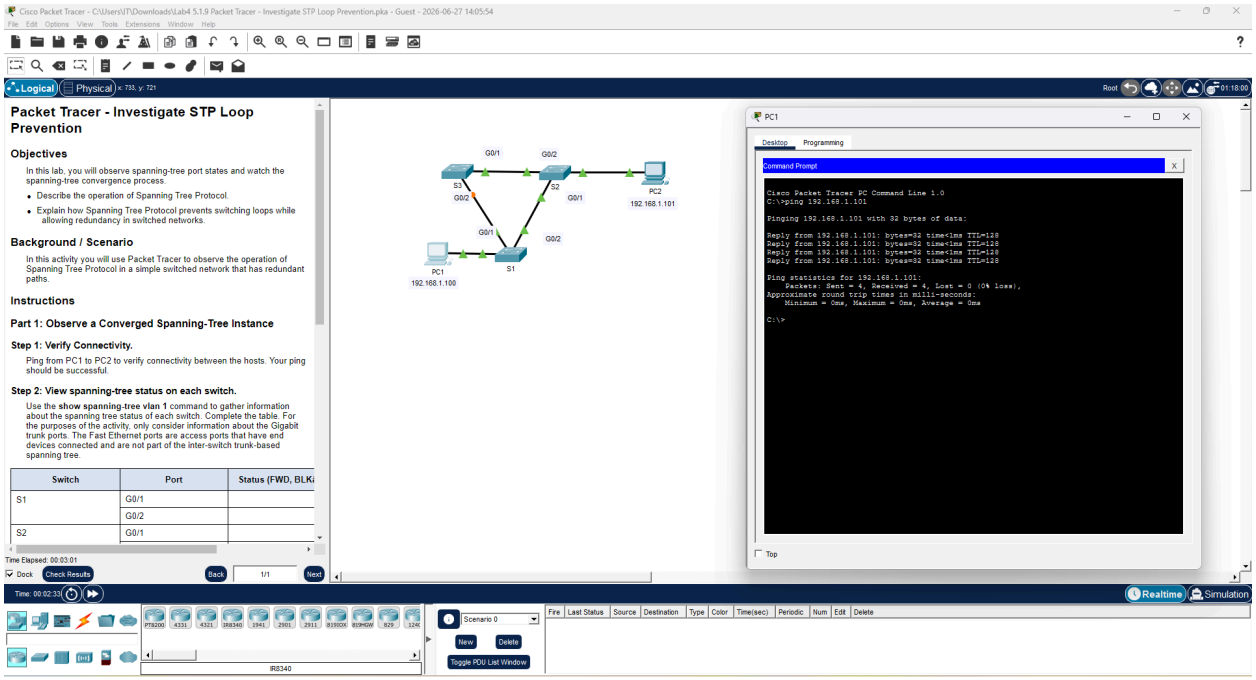
```
None
S2> enable
S2# configure terminal
S2(config)# interface range g0/1 - 2
S2(config-if-range)# switchport mode trunk
S2(config-if-range)# no shutdown
S2(config-if-range)# exit
S2(config)# interface FastEthernet0/1
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 1
S2(config-if)# no shutdown
S2(config-if)# exit
S2(config)# interface vlan 1
S2(config-if)# ip address 192.168.1.12 255.255.255.0
S2(config-if)# no shutdown
S2(config-if)# exit
```

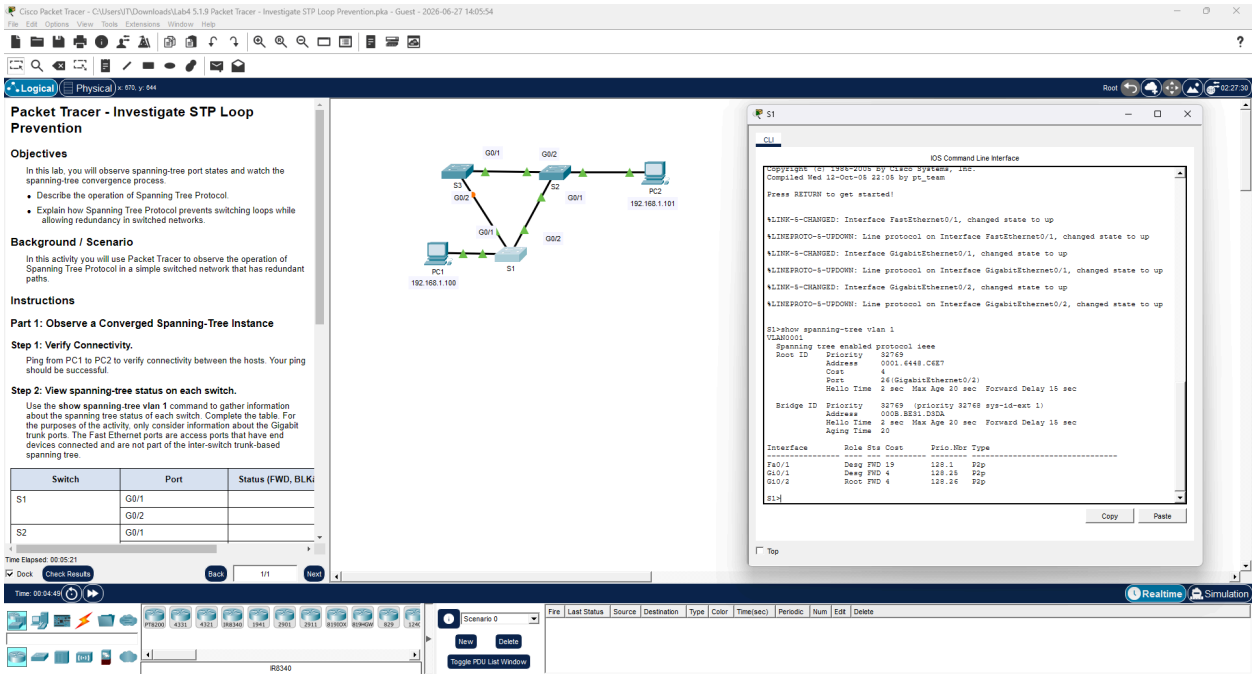
Switch 3 (S3)

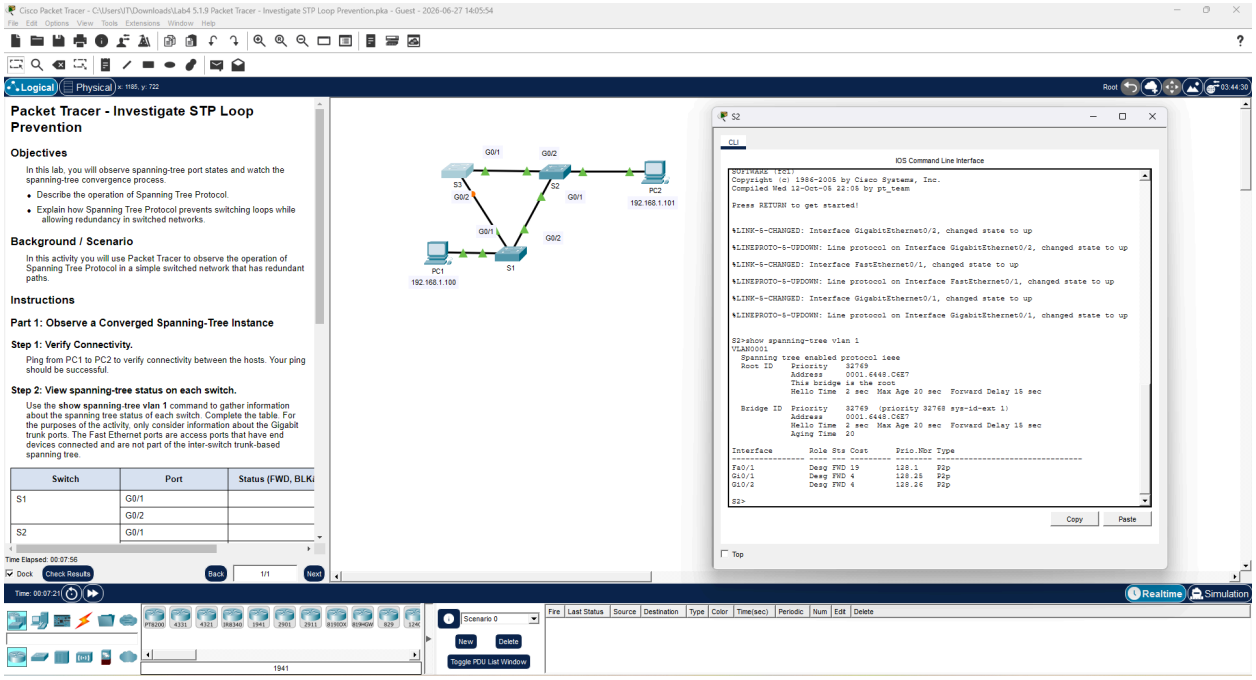
```
None
S3> enable
```

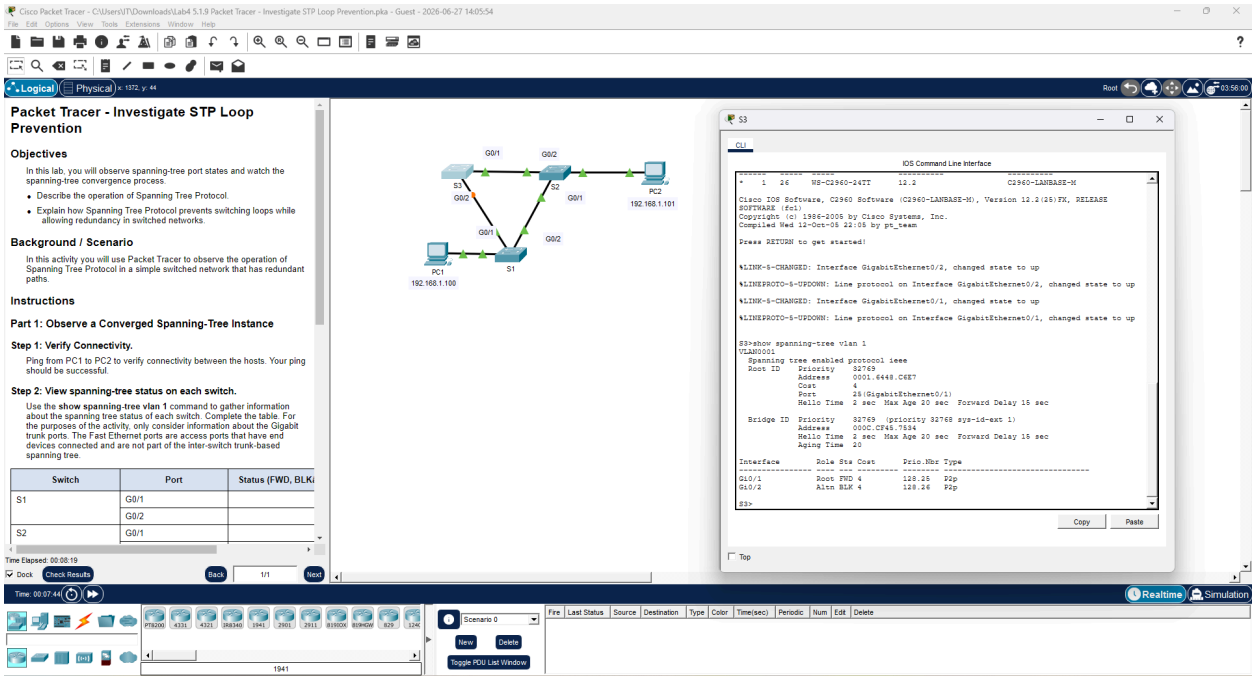
```
S3# configure terminal
S3(config)# interface range g0/1 - 2
S3(config-if-range)# switchport mode trunk
S3(config-if-range)# no shutdown
S3(config-if-range)# exit
S3(config)# interface FastEthernet0/1
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 1
S3(config-if)# no shutdown
S3(config-if)# exit
S3(config)# interface vlan 1
S3(config-if)# ip address 192.168.1.13 255.255.255.0
S3(config-if)# no shutdown
S3(config-if)# exit
```

5. Spanning Tree State Analysis




Cisco Packet Tracer - C:\Users\IT\Downloads\Lab4 5.1.9 Packet Tracer - Investigate STP Loop Prevention.pka - Guest - 2026-06-27 14:05:54
Logical
Physical
Packet Tracer - Investigate STP Loop Prevention
Objectives
In this lab, you will observe spanning-tree port states and watch the spanning-tree convergence process.
Describe the operation of Spanning Tree Protocol.
Explain how Spanning Tree Protocol prevents switching loops while allowing redundancy in switched networks.
Background / Scenario
In this activity you will use Packet Tracer to observe the operation of Spanning Tree Protocol in a simple switched network that has redundant paths.
Instructions
Part 1: Observe a Converged Spanning-Tree Instance
Step 1: Verify Connectivity.
Ping from PC1 to PC2 to verify connectivity between the hosts. Your ping should be successful.
Step 2: View spanning-tree status on each switch.
Use the show spanning-tree vlan 1 command to gather information about the spanning tree status of each switch. Complete the table. For the purposes of the activity, only consider information about the Gigabit trunk ports. The Fast Ethernet ports are access ports that have end devices connected and are not part of the inter-switch trunk-based spanning tree.
Switch
Port
Status (FWD, BLK
S1
G0/1
G0/2
S2
G0/1
Time Elapsed: 00:05:21
Dock
Check Results
Back
1/1
Next
G0/1
G0/2
S3
S2
PC2
192.168.1.101
PC1
192.168.1.100
S1
S1
CLI
IOS Command Line Interface
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
S1>show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 26(GigabitEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000B.BE31.D5DA
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/1 Desg FWD 4 128.25 P2p
Gi0/2 Root FWD 4 128.26 P2p
S1>
Copy
Paste
Top
Time: 00:04:49
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
IR8340


Cisco Packet Tracer - C:\Users\IT\Downloads\Lab4 5.1.9 Packet Tracer - Investigate STP Loop Prevention.pka - Guest - 2026-06-27 14:05:54
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Switch
Port
Status (FWD, BLK
S1
G0/1
G0/2
S2
G0/1
Time Elapsed: 00:07:56
Dock
Check Results
Back
1/1
Next
G0/1
G0/2
S3
S2
PC2
192.168.1.101
PC1
192.168.1.100
S1
S2
CLI
IOS Command Line Interface
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team
Press RETURN to get started!
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
S2>show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0001.6448.C6E7
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/1 Desg FWD 4 128.25 P2p
Gi0/2 Desg FWD 4 128.26 P2p
S2>
Copy
Paste
Top
Time: 00:07:21
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
1941



Part 1: Converged Instance State Table

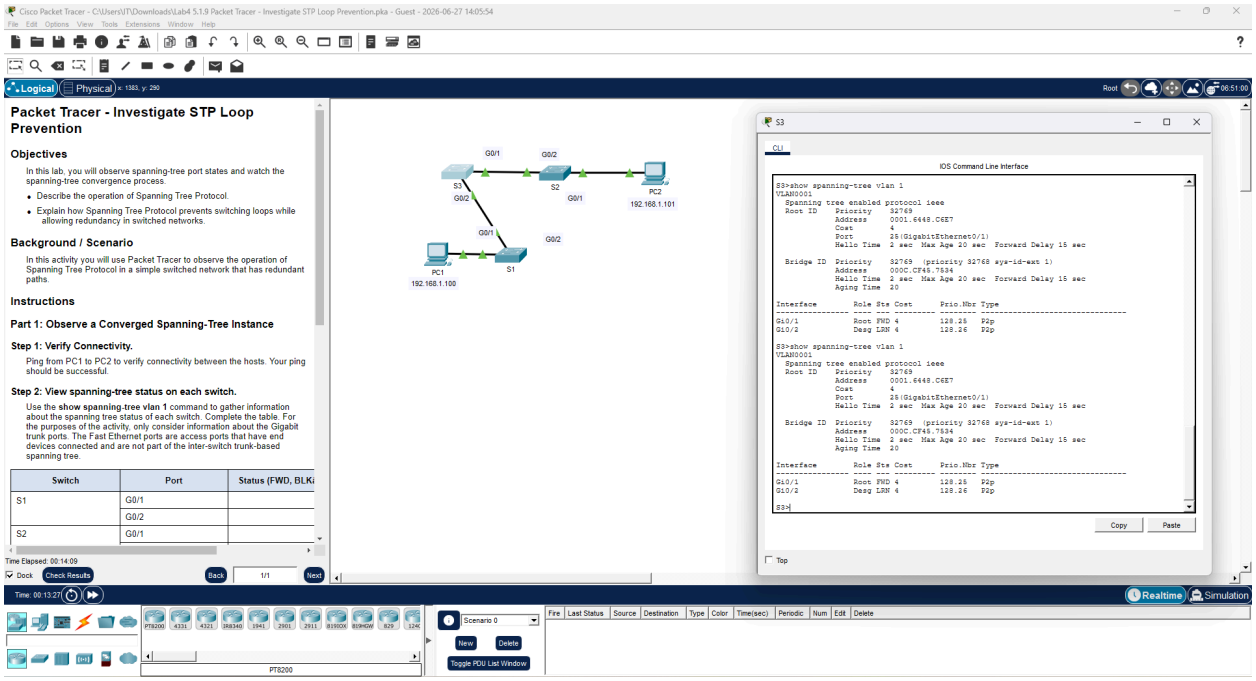
The following table reflects the spanning tree status of each Gigabit trunk port in a stable environment.

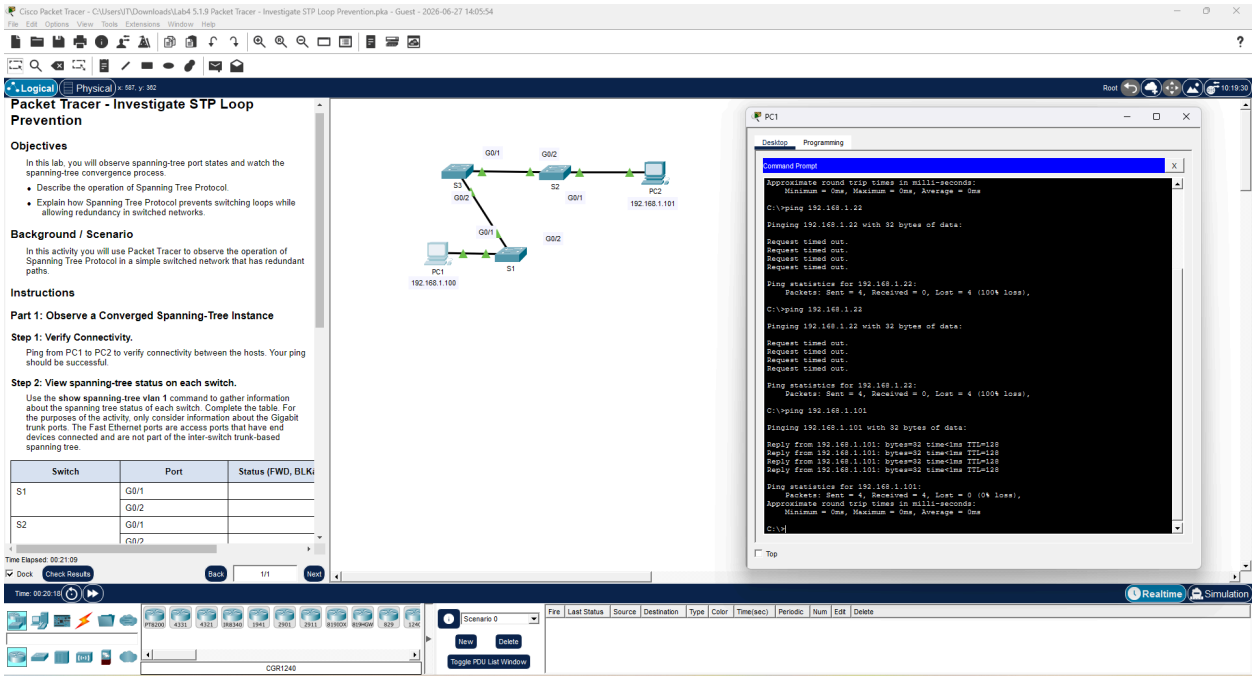
Switch	Port	Status	Root Bridge?
S1	G0/1	FWD	Yes
S1	G0/2	FWD	Yes
S2	G0/1	FWD	No
S2	G0/2	FWD	No
S3	G0/1	FWD	No
S3	G0/2	BLK	No

Guided Questions

- **What does the orange link light mean?** It indicates that the port is in a non-forwarding state, specifically the Blocking (BLK) state enforced by STP.
- **What path will frames take from PC1 to PC2?** Frames travel from PC1 $\rightarrow$ S2 $\rightarrow$ S1 $\rightarrow$ S3 $\rightarrow$ PC2.
- **Why do the frames not travel through S3 directly?** The direct path between S2 and S3 contains the port placed in a blocking state (G0/2 on S3) to eliminate the logical loop.
- **Why has spanning tree placed a port in a blocking state?** To prevent data frames from continuously looping around the redundant triangle path, which would cause a broadcast storm.

6. Observed Convergence





Convergence Process

When the cable connecting S1 and S2 is removed, the network must reconverge. Port G0/2 on S3 transitions through the following states:

1. **BLK (Blocking):** Drops data traffic.
2. **LIS (Listening):** Processes BPDUs to ensure no loop exists.
3. **LRN (Learning):** Builds the MAC address table.
4. **FWD (Forwarding):** Becomes fully operational for data frames.

Post-Convergence Observations

- **Are any ports showing an orange link light after removing the cable? Why or why not?** No, there are no longer any orange link lights because the physical loop has been broken by the removal of the cable, eliminating the need for STP to block any ports.

7. Reflections

This activity demonstrates that STP is an essential protocol for maintaining network stability by automatically managing link changes without manual intervention. However, the observation of the Listening and Learning phases highlights the inherent slowness of legacy STP (802.1D), which can take up to 30–50 seconds to fail over, emphasizing the importance of Rapid Spanning Tree Protocol (RSTP) in modern environments.